Graph-Based Risk Mapping: Mining Road Networks to Predict Vehicular Mortality in Florida Panthers

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Abstract

Appears first, written last.

Introduction

Motivation

The Florida panther (*Puma concolor coryi*) has long been listed as federally endangered, but conservation efforts have allowed its population to grow over the past two decades. As panthers expand into an increasingly developed landscape, however, vehicle collisions have become a major threat, with documented roadkills ranging from 6 to 34 per year since 2000 (Florida Fish and Wildlife Conservation Commission 2025). For a population estimated at roughly 120 to 230 adults (Florida Fish and Wildlife Conservation Commission 2025), losses at this scale are significant, with vehicle collisions accounting for roughly one in five deaths or serious injuries among radio-collared individuals (Prat-Guitart et al. 2026).

These collisions rose alongside the population's recovery and have remained persistently high, with occurrence probabilities increasing during the early years of recent monitoring and then stabilizing (Prat-Guitart et al. 2026). The collisions are also geographically concentrated, with 92.3% of recorded vehicular deaths occurring in four counties (Collier, Hendry, Lee, and Glades) (Florida Fish and Wildlife Conservation Commission nd), thus, becoming our regional scope of study.

Wildlife crossing structures paired with fencing have sharply reduced deaths where they have been installed (Florida Fish and Wildlife Conservation Commission 2025), but they are expensive, and limited funding restricts how many can be built (Loraamm and Downs 2016). Agencies must therefore decide which stretches of road to prioritize, which requires a reliable way to rank road segments by their risk of a fatal collision.

This paper attempts to address the following research question: *Can a graph neural network trained on telemetry-derived movement connectivity outperform non-spatial baseline models in predicting road segments at high risk of Florida panther vehicular mortality?*

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We hypothesize that road segments linked by shared panther movement share collision risk, and that a GNN trained on this connectivity will rank high-risk segments more accurately than non-spatial baselines using the same features. To test this, we construct a road-segment graph for the selected four-county study area that combines FDOT physical adjacency with movement-derived edges from over 63,000 panther telemetry fixes, train a GraphSAGE model on this graph, and compare it against random forest, XGBoost, and kernel density baselines on identical features and identical spatial cross-validation folds. Because collisions are rare, we evaluate ranking performance with AUC-PR and precision-at-K, and assess calibration with the Brier score.

Related Works

Research surrounding Florida Panther populations and their interactions with anthropologic factors has spanned many different disciplines, seeking to explain, mitigate their impacts, and prevent future mortalities. Prior work has approached this problem from three directions, while no prior work attempts to combine them.

Spatial Patterns of Vehicular Mortality

Statistical and machine learning models for wildlife–vehicle collisions predict risk from traffic, landscape, and habitat features (Pagany 2022; Laube et al. 2023; Visintin, van der Ree, and McCarthy 2016; Bell, Wang, and Ament 2024). However, these models evaluate each road segment from its own attributes without representing connectivity between segments, and they estimate animal presence indirectly, through habitat suitability (Visintin, van der Ree, and McCarthy 2016) or land-cover neighborhood features (Laube et al. 2023). None incorporate individual movement data to examine the paths animals take through the landscape and why those paths converge on particular road segments. While Prat-Guitart et al. (2026) used dynamic occupancy models to estimate probabilities of collision occurrence on road segments while accounting for imperfect detection, modeling occupancy at the segment level means the paths panthers take to reach those segments are not represented.

Panther Movement and Landscape Factors

Panther-specific studies find collisions concentrate on segments with connected landscapes and low resistance to panther movement (Schwab and Zandbergen 2011; U.S. Fish and Wildlife Service 2019; Prat-Guitart et al. 2026), but infer movement potential from landscape resistance rather than from observed telemetry. Panther telemetry has been used to site crossings, where tracking-based siting selects different locations than mortality-based siting (Loraamm and Downs 2016; Downs et al. 2014), suggesting movement data captures risk that collision records miss. Graph neural networks (GNNs) have improved road-segment accident prediction by learning across connections between segments (Nippani et al. 2023; Yu et al. 2021), but only for human traffic. We hypothesize that road segments linked by shared panther movement also share collision risk, and that a GNN can exploit this structure.

Mitigation and Crossing Placement

Among the most effective measures for reducing Florida panther-vehicle mortality, we have wildlife crossings and associated fencing. However, higher construction costs make strategic placement necessary. The Florida Fish and Wildlife Conservation Commission reports that panther deaths been sharply reduced along road sections where crossings and fencing are present, while earlier interventions such as reduced nighttime speed limits or reflectors did not eliminate collisions (Florida Fish and Wildlife Conservation Commission 2025). Previous study has approached crossing placement based on both mortality records and movement data, applying maximal covering location model to identify candidate crossing locations using either historical vehicle mortality locations or radio telemetry observations (Downs et al. 2014). This approach was later extended by estimating probable animal-road interactions based on telemetry and optimizing crossing placement under a limited number of available structures and fencing distances (Loraamm and Downs 2016). Similarly, the U.S. fish and Wildlife Service identifies road hot spots from documented collisions while recommending that land cover, telemetry and habitat connectivity also inform final crossing placement location (U.S. Fish and Wildlife Service 2019). These studies show is the value of combining both mortality and movement information for mitigation planning; however, they primarily treat crossing placement as a spatial optimization problem. Our approach instead aims to address the preceding prediction problem, which is learning whether connectivity between road segments derived from observed panther movement could improve the ranking of segments by future mortality risk.

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AI Usage Statement

During the preparation of this work, the authors used Anthropic's Claude to assist in rewording text. All ideas and literature analysis are the authors' own. The authors reviewed and edited the output and take full responsibility for the content of this publication.